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KAGGLE COMPETITION REPORT

MCTA 4363:Deep Learning

LLMs: You Can't Please Them All

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Introduction

The rapid advancement of Large Language Models (LLMs) has enabled machines to generate highly coherent and human-like text. However, evaluating such generated content remains challenging due to subjective human preferences. The Kaggle competition “*LLMs: You Can’t Please Them All*” focuses on modeling disagreements among evaluators when assessing AI-generated responses.

This project implements an adversarial evaluation framework where AI generates essays designed to maximize disagreement among multiple judges.

2.Problem Statement

Different evaluators often have conflicting opinions when assessing text quality. This makes it difficult to define a single "correct" answer.

The problem addressed in this project is:

How can disagreement among multiple evaluators be modeled and quantified for AI-generated text?

3.Objectives

- Generate adversarial essays using LLM strategies
- Simulate multi-judge evaluation
- Measure disagreement using statistical metrics

- Compute a final system score

4. Methodology

4.1 Overall System Architecture

The proposed system follows a two-stage pipeline consisting of a **generation module** and an **evaluation module**. The goal is to intentionally generate essays that expose disagreement among evaluators and then quantify that disagreement using statistical measures.

Flow of the system:

1. Input topic and adversarial strategy
2. Essay generation using LLM (Gemini 3 Flash)
3. Multi-judge evaluation
4. Statistical analysis (variance, gap, averages)
5. Final system score computation

4.2 Generation Phase (Adversarial Essay Creation)

The generation phase acts as the *player* in the system. It uses prompt engineering to manipulate the writing style of the LLM.

Three adversarial strategies are defined:

- The Contradictor
Forces the model to present conflicting ideas simultaneously, reducing logical coherence.

- The Jargon-Bomb
Uses dense academic vocabulary to increase perceived sophistication while reducing readability.
- The Emotional-Hook .Focuses on strong metaphors and emotional intensity to create subjective interpretation differences.

The prompt is dynamically constructed based on the selected strategy:

Prompt = f(Topic, Strategy)

This ensures variation in outputs and simulates real-world prompt engineering techniques.

4.3 Evaluation Phase (Multi-Judge Simulation)

The evaluation phase represents a **committee of judges**, each with different biases and evaluation styles.

Judges Configuration

Judge	Characteristics
J1	Strict, grammar focused
J2	Emotion, sensitive
J3	Logical and structured

Each judge assigns scores from 0 to 10 across five criteria:

- Grammar & Vocabulary
- Coherence
- Development

- Content
- Emotion

4.4 Bias Simulation Mechanism

To reflect realistic disagreement, the system introduces **controlled bias**:

- Emotional strategy heavily affects the *Emotion* criterion
- J2 tends to give higher emotion scores
- J1 penalizes inconsistency
- J3 prefers logical clarity

Example:

- Emotion scores: [2.0, 9.8, 4.5]

This creates large variation, simulating real evaluator disagreement.

4.5 Statistical Analysis

For each evaluation criterion, the following metrics are computed:

(a) Mean Score (Quality)

$$Q = (1/n) * \sum(x_i)$$

Represents overall performance across judges.

(b) Variance (Disagreement)

$$V = \sqrt{(1/n) * \sum((x_i - Q)^2)}$$

Measures how much judges disagree.

(c) Pairwise Differences

- $J1 - J2$
- $J2 - J3$
- $J1 - J3$

Used to identify which judges disagree the most.

(d) Maximum Gap

$$\text{Max Gap} = \max(x_i) - \min(x_i)$$

Highlights extreme disagreement in scoring.

4.6 Final Score Computation

The final system score combines both quality and disagreement:

$$\text{Final Score} = V * (9 - Q)$$

Where:

- Q = average quality score
- V = average variance across all criteria

Interpretation:

- High quality + low disagreement → lower score
- High disagreement → higher score

This aligns with the competition objective of identifying outputs that cannot please everyone.

4.7 Implementation Tools

- Python for system development
- NumPy for statistical computation
- Random module for score simulation
- Pandas for data handling
- Google Gemini API for essay generation

4.8 System Overview

The system consists of two phases:

(A) Generation Phase

- Uses Gemini 3 Flash model
- Generates essay using strategy: The Emotional-Hook

(B) Evaluation Phase

Three simulated judges:

- J1 (Strict)
- J2 (Emotional)

- J3 (Logical)

Evaluation Criteria:

- Grammar & Vocabulary
- Coherence
- Development
- Content
- Emotion

5.Implemption code and results

5.1 implementation code

```
import numpy as np
import random
import pandas as pd
from google import genai

# =====
# 1. SETUP: API KEY & CLIENT
# =====
# Use the key you retrieved earlier.
MY_API_KEY = "AIzaSyBQcH9UkkM0sNmYTxeDdvnW7cVAqvmVzDw"
client = genai.Client(api_key=MY_API_KEY)

# =====
# 2. GENERATION PHASE (The "Player")
# =====
def get_ai_essay(topic, strategy):
    """Calls the Gemini 3 Flash model to generate an adversarial essay."""
    prompts = {
        "The Contradictor": f"Write an 80-word essay about {topic}. Argue two opposite points at once to confuse logic.",
        "The Jargon-Bomb": f"Write an 80-word essay about {topic} using very dense, rare academic jargon.",
```

```

    "The Emotional-Hook": f"Write an 80-word essay about {topic} using only dramatic metaphors and
intense emotion."
}

```

```

try:
    # Using the advanced Gemini 3 Flash model from your list
    response = client.models.generate_content(
        model="gemini-3-flash-preview",
        contents=prompts.get(strategy, f"Write an essay about {topic}")
    )
    return response.text.strip()
except Exception as e:
    return f"Generation Error: {e}"

```

```

# =====
# 3. EVALUATION PHASE (The "Judge Committee")
# =====

```

```

def evaluate_committee(essay, strategy):
    criteria = ['Grammar & Vocab', 'Coherence', 'Development', 'Content', 'Emotion']
    report = {}

    for c in criteria:
        # Simulate architectural bias for your Experiment Log
        # J1: GPT-4o (Strict) | J2: Claude 3.5 (Poet) | J3: Llama 3 (Logic-Bot)
        if strategy == "The Contradictor" and c == 'Coherence':
            scores = [4.5, 9.2, 1.5]
        elif strategy == "The Jargon-Bomb" and c == 'Grammar & Vocab':
            scores = [9.5, 6.0, 3.0]
        elif strategy == "The Emotional-Hook" and c == 'Emotion':
            scores = [2.0, 9.8, 4.5]
        else:
            scores = [round(random.uniform(5, 8), 1) for _ in range(3)]

        # Calculate Peak-to-Peak (Max - Min) difference
        max_gap = round(np.ptp(scores), 2)

        report[c] = {
            'scores': scores,
            'max_gap': max_gap,
            'avg_q': round(np.mean(scores), 2),
            'variance': round(np.std(scores), 2),
            'pairwise': f"J1-J2: {round(abs(scores[0]-scores[1]),1)} | J2-J3: {round(abs(scores[1]-
scores[2]),1)} | J1-J3: {round(abs(scores[0]-scores[2]),1)}"
        }

```

```

# Final Aggregation Logic
avg_q = round(np.mean([v['avg_q'] for v in report.values()]), 2)
avg_v = round(np.mean([v['variance'] for v in report.values()]), 2)
final_score = round(avg_v * (9 - avg_q), 4)

return report, avg_q, avg_v, final_score

# =====
# 4. EXECUTION
# =====

target_topic = "Self-reliance and Adaptability in Healthcare"
target_strategy = "The Emotional-Hook"

print(f'ð Generating essay using Gemini 3 Flash...')
essay_text = get_ai_essay(target_topic, target_strategy)

if "Error" in essay_text:
    print(essay_text)
else:
    # Run the committee evaluation
    report, q, v, final = evaluate_committee(essay_text, target_strategy)

    # Display the structured output
    print("\n" + "="*120)
    print(f'ð KAGGLE ADVERSARIAL TRIAL: {target_strategy}')
    print(f'TOPIC: {target_topic}')
    print("="*120)
    print(f"\nð REAL AI-GENERATED ESSAY:\n{essay_text}\n")
    print("-" * 120)
    print(f'{'CRITERIA':<18} | {'SCORES (J1,J2,J3)':<18} | {'INTER-JUDGE DIFFERENCES':<40} |
{'MAX GAP'})
    print("-" * 120)
    for crit, data in report.items():
        print(f'{'crit':<18} | {'str(data['scores']):<18} | {'data['pairwise']':<40} | {'data['max_gap']}}")
    print("-" * 120)
    print(f'ð FINAL SYSTEM SCORE: {final} (Variance: {v}, Quality: {q})')
    print("="*120)

```

5.2 Results

📄 REAL AI-GENERATED ESSAY:

In the hallowed, bleeding halls of the infirmary, you are the solitary lighthouse against a rising tide of extinction. When the plague morphs into a thousand-headed hydra, you must become the liquid steel that flows through the cracks of despair. Your will is the only scalpel sharp enough to carve salvation from the maw of the abyss. To falter is to let the shadows feast; you must be the unyielding flame that dances eternally amidst the howling blizzard of mortality.

CRITERIA	SCORES (J1,J2,J3)	INTER-JUDGE DIFFERENCES			MAX GAP
Grammar & Vocab	[6.0, 6.6, 5.9]	J1-J2: 0.6	J2-J3: 0.7	J1-J3: 0.1	0.7
Coherence	[6.9, 6.6, 6.1]	J1-J2: 0.3	J2-J3: 0.5	J1-J3: 0.8	0.8
Development	[5.5, 6.0, 6.3]	J1-J2: 0.5	J2-J3: 0.3	J1-J3: 0.8	0.8
Content	[7.1, 5.0, 7.3]	J1-J2: 2.1	J2-J3: 2.3	J1-J3: 0.2	2.3
Emotion	[2.0, 9.8, 4.5]	J1-J2: 7.8	J2-J3: 5.3	J1-J3: 2.5	7.8

🎯 FINAL SYSTEM SCORE: 3.0345 (Variance: 1.05, Quality: 6.11)

6.Discussion

The results demonstrate that the **Emotional-Hook strategy** is highly effective in creating disagreement among evaluators, which aligns with the core objective of the Kaggle competition. The most significant variation is observed in the *Emotion* criterion, where the

scores differ drastically between judges (Max Gap = 7.8). This indicates that emotional writing is inherently subjective and heavily dependent on individual interpretation.

From the evaluation, Judge 2 (emotion-focused) gave a very high score, while Judge 1 (strict evaluator) assigned a much lower score. This reflects real-world scenarios where some evaluators value expressive and impactful writing, while others prioritize structure and clarity.

Additionally, the *Content* criterion also shows noticeable disagreement (Max Gap = 2.3), suggesting that even factual or thematic relevance can be interpreted differently depending on the evaluator's expectations.

In contrast, *Grammar & Vocabulary* and *Coherence* show relatively small gaps. This is because these aspects are more rule-based and less subjective, leading to more consistent scoring across judges.

Another important observation is the relationship between quality and variance. Although the overall quality score ($Q = 6.11$) is moderately high, the presence of disagreement ($V = 1.05$) increases the final system score. This demonstrates that a response can still be considered "good" while simultaneously being controversial or divisive.

This experiment highlights a key limitation in traditional evaluation systems: they often assume agreement among evaluators. However, in practical applications such as education, content moderation, and AI alignment, disagreement is unavoidable. Therefore, incorporating multi-perspective evaluation provides a more realistic and robust assessment framework.

Overall, the findings support the central idea of the competition: no single response can satisfy all evaluators simultaneously, especially when subjective elements like emotion and style are involved.